

MOHAMED RAGAB

Backend / Laravel Developer | PHP | REST APIs | MySQL | Full-Stack

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PROFESSIONAL SUMMARY

Backend-focused software developer specializing in PHP and Laravel, with hands-on experience designing REST APIs, relational databases, authentication and authorization systems, and full-stack web applications. Built and maintain a full-stack healthcare platform (Shifago) covering backend services, real-time features, and dashboard development, and currently working as a Backend Developer on a healthcare platform (ADHD Compass). Designed and implemented the Laravel backend, database architecture, and vector-storage infrastructure for GlucoRAG, a medical RAG (Retrieval-Augmented Generation) application developed during AI Hackathon 2026, gaining practical exposure to FastAPI, LangChain, and Supabase/pgvector. Electrical Engineering (Electronics & Communications) student with strong analytical and problem-solving skills.

TECHNICAL SKILLS

Programming Languages: PHP, JavaScript, TypeScript, Python

Backend: Laravel, REST APIs, FastAPI

Frontend: Blade, JavaScript, Axios, Bootstrap, React, TypeScript, Tailwind CSS

Databases: MySQL, PostgreSQL, Supabase, pgvector

Authentication & Security: Laravel Sanctum, Authentication, Authorization, Middleware, RBAC, Rate Limiting

Real-Time: Laravel Reverb

AI / RAG: Retrieval-Augmented Generation (RAG), LangChain, Embeddings, Vector Search, Semantic Retrieval, LLM Integration, Citation & Grounding

Tools: Git, GitHub, Linux

AI-Assisted Development: Google Antigravity

PROFESSIONAL EXPERIENCE

Backend Developer — ADHD Compass

June 2026 – Present

Cairo, Egypt

- Developing backend functionality for a healthcare platform focused on providing medical consultation and support services related to ADHD.

Full-Stack Laravel Developer — Shifago

October 2025 – Present

Egypt | shifago.health | Healthcare / home medical care platform

- Built and maintain a full-stack healthcare platform end-to-end using Laravel, PHP, MySQL, Blade, JavaScript, Bootstrap, and Axios, covering both backend services and frontend interfaces.
- Designed and implemented backend REST APIs, database schema, and dashboard functionality supporting healthcare service workflows.
- Implemented authentication and authorization using Laravel Sanctum, and real-time features using Laravel Reverb.
- Built and integrated frontend interfaces with Blade, JavaScript, and Axios, connecting them to backend APIs.
- Optimized database queries and application performance, and collaborated using Git/GitHub following Laravel best practices and clean architecture.
- Diagnosed and resolved application issues while continuously delivering new features and maintainability improvements.

SELECTED PROJECTS

GlucoRAG — Medical RAG Application (Type 2 Diabetes Screening Assistant)

AI Hackathon 2026 Project | Laravel, PHP, MySQL, Sanctum, FastAPI, LangChain, Supabase/pgvector, React, TypeScript

- Developed during AI Hackathon 2026 as a Type 2 Diabetes screening-focused medical RAG (Retrieval-Augmented Generation) application.
- Designed and implemented the Laravel backend architecture, including REST APIs, Sanctum-based authentication and authorization, conversation/message persistence, citation storage, and asynchronous queue-based job processing that integrates with an external RAG service.
- Designed and implemented the application database schema (users, conversations, messages, message citations), including relationships, foreign key constraints, indexes, and cascading delete behavior enforcing per-user data ownership.
- Designed and implemented the Supabase/pgvector vector-storage infrastructure, separate from the application database, supporting document ingestion, chunking, embedding storage, and similarity-based retrieval.
- Architected the overall system connecting a React/TypeScript frontend, the Laravel API, a FastAPI + LangChain RAG service, and Supabase/pgvector; the FastAPI RAG service, LangChain pipeline, React frontend, and evaluation/safety layer were developed with the assistance of Google Antigravity (AI-assisted development).
- Implemented evaluation and safety validation covering retrieval quality, citation integrity, groundedness, and fail-closed handling of out-of-scope or unsupported medical requests (e.g., diagnosis, treatment, and emergency queries).

E-Commerce RESTful API

Laravel, PHP, MySQL, Sanctum

- Built a Laravel RESTful API with role-based access control (Admin, Seller, Customer roles), product and category management, shopping cart, and order management, using a layered architecture (Controllers → Form Requests → Services → Models).
- Implemented Laravel Sanctum authentication, event/listener-based email notifications, form request validation, model factories, and database seeders.

Taskaty — Task Management Application

- Developed a task management application featuring authentication, CRUD task operations, categorization, priority levels, and status tracking.

EDUCATION

Al-Azhar University

October 2023 – July 2028

Bachelor's Degree, Electrical Engineering (Electronics & Communications)

HACKATHONS & ACTIVITIES

AI Hackathon 2026

August 2026

Organized in partnership with CREATIVA Innovation Hubs, Orange Digital Center Egypt, ITIDA, TIEC, and INSTANT Software Solutions

- Selected to participate in AI Hackathon 2026 (August 16–20, 2026), a hands-on AI development program focused on large language models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, and vector databases, applying these concepts to build reliable, practical AI systems.
- Developed GlucoRAG, a Type 2 Diabetes screening-focused medical RAG application, as the hands-on project for the event (see Selected Projects).

LEADERSHIP / ADDITIONAL EXPERIENCE

President, Hult Prize Al-Azhar University Team

October 2025 – February 2026

Campus Director (HR & Organizing Committee), Hult Prize Al-Azhar

June 2025 – October 2025

- Led HR recruitment and team development, and organized campus events and workshops.

CERTIFICATIONS / TRAINING

- Vital Training Program — 92% (5th in Governorate)
- Participation Certificate, Tech Nights Bootcamp

- Student Conference Participation — Faculty of Engineering (Electrical Department)
- AI Image Generation
- Content Writing

LANGUAGES

Arabic (Native or Bilingual) | English (Professional Working) | German (Limited Working)